Tiered Lambda Architecture for Real-Time and Batch Data in Healthcare System's

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Abstract— Alarm fatigue in intensive care units (ICUs) constitutes a major patient-safety challenge, with contemporary monitoring systems reporting false-positive alarm rates between 80% and 99%, leading to clinician desensitization, delayed responses, and increased operational costs. A primary cause of this problem is the architectural fragmentation of healthcare data pipelines, in which real-time physiological streams are processed independently from historical patient records and clinical context, resulting in alert generation based on incomplete and low-quality information. In this study, we propose a Tiered Lambda Architecture that integrates real-time and batch processing while introducing multi-layer data quality validation prior to machine-learning-based inference.The proposed framework consists of three layers: (i) a Speed Layer that performs physiological and temporal validation to eliminate sensor artifacts and malformed data in sub-millisecond time, (ii) a Hybrid Layer that applies contextual anomaly detection using an Isolation Forest model under strict latency constraints, and (iii) a Serving Layer that unifies real-time alerts with historical electronic health record data to provide clinicians with a coherent and queryable patient view. The system was evaluated using 5,000 heart-rate records sampled from the MIMIC-III clinical database and compared against a conventional rule-based threshold monitoring system representative of standard ICU practice.Experimental results demonstrate that early-stage data quality filtering alone removes 25.9% of corrupted sensor records, preventing noise propagation into downstream analytics. Overall, the proposed architecture reduced total false alarms by 37.7% (from 310 to 193), lowered the false-positive rate from 92.6% to 46.1%, and increased precision from 0.074 to 0.539 while maintaining perfect recall (1.0), ensuring that no true clinical anomalies were missed. End-to-end processing latency averaged 0.54 ms per record, satisfying real-time clinical requirements and outperforming batch-based baselines by more than three orders of magnitude.These findings indicate that architectural design choices—specifically the prioritization of data quality and contextual integration before model complexity—play a more decisive role in mitigating alarm fatigue than the exclusive adoption of advanced learning algorithms. The proposed Tiered Lambda Architecture provides a scalable, regulatory-aware foundation for next-generation ICU monitoring systems that reduce cognitive burden on healthcare staff without compromising patient safety.

I. INTRODUCTION

Intensive care units (ICUs) have become heavily dependent on continuous physiological monitoring systems to support timely clinical decision-making and patient safety. A single monitored bed generates approximately 43,200 vital-sign measurements during a 12-hour shift when sampled at 1 Hz, producing a large and rapidly growing stream of real-time data [1], [8]. Contemporary ICU monitoring platforms rely primarily on simple statistical threshold rules, such as triggering heart-rate alarms when values exceed 120 bpm or fall below 50 bpm [15], [16]. Although computationally efficient, this approach results in an excessive number of alarms, the majority of which are clinically non-actionable.

The persistent generation of spurious alerts has led to the widespread phenomenon known as alarm fatigue, which significantly undermines the effectiveness of clinical monitoring systems. Prior studies report that 23% of delayed diagnoses in ICUs are directly attributable to clinician desensitization caused by frequent false alarms [4], while nursing staff spend approximately 25–30% of their working shifts responding to and investigating alarms that do not require intervention [15]. Beyond patient safety implications, false alarms impose measurable financial costs, estimated at $0.071 per alarm in nursing labor alone based on analysis of the MIMIC-III database [10]. Collectively, these factors demonstrate that excessive alerting is not merely an inconvenience but a systemic clinical and economic burden.

Despite extensive research into improving alarm accuracy, the fundamental cause of this problem is not solely algorithmic but architectural in nature. Current real-time monitoring systems operate largely in isolation from critical contextual information, including patient-specific baseline physiology, active medications, diagnostic history, long-term physiological trends, and real-time data quality indicators [3], [18], [20]. As a consequence, alerts are generated based on incomplete and potentially corrupted sensor data, forcing rule-based systems to operate without clinical context and resulting in false-positive rates as high as 80–99% [5], [11].

Recent advances have explored multiple directions to address this limitation. Machine-learning-based approaches, including LSTM autoencoders and generative models [7], [11], achieve high predictive accuracy on retrospective datasets such as MIMIC-III [10]; however, they require labeled training data, introduce substantial inference latency, and remain highly sensitive to noisy inputs, thereby propagating sensor artifacts directly into clinical predictions [2], [12]. Data warehousing systems enable comprehensive historical analysis [6], [18], but their batch-oriented

processing introduces delays of 12–24 hours, rendering them unsuitable for real-time clinical alerting. Similarly, real-time stream-processing frameworks such as Apache Kafka and Spark-inspired edge processing systems [1], [8] provide scalable ingestion and computation but lack unified clinical context models and do not incorporate systematic data quality validation prior to alert generation [3], [19].

To address these limitations, this study proposes a tiered lambda architecture for ICU monitoring systems that integrates real-time and batch data processing while prioritizing data quality and contextual awareness. The proposed framework introduces:

1. Immediate physiological data validation to eliminate sensor noise at ingestion time [9], [19],
2. Contextual anomaly detection using lightweight machine-learning inference.
3. A unified serving layer that merges real-time alerts with historical electronic health records for clinical interpretation.

II. LITERATURE REVIEW

Alarm fatigue has been widely recognized as a critical challenge in intensive care units, particularly with the increasing reliance on continuous physiological monitoring systems for patient safety and clinical decision support [4]. Modern ICUs generate massive volumes of high-frequency vital-sign data, which are processed by rule-based alerting systems to detect potential clinical deterioration [1], [8]. While these systems aim to ensure early intervention, their limited contextual awareness and sensitivity to sensor noise have resulted in excessive false alarms, reducing clinical trust and increasing cognitive workload [4], [5]. Consequently, researchers have explored various algorithmic, statistical, and architectural strategies to improve alarm reliability, reduce false positives, and enhance real-time clinical interpretation [11], [12], [14]. This section provides a critical overview of existing approaches, their strengths and limitations, and the unresolved challenges that motivate the proposed solution.

Early ICU alarm systems were primarily based on static threshold rules derived from population-level physiological ranges [15], [16]. These systems evaluated individual vital signs independently and triggered alerts whenever values crossed predefined boundaries. Although such approaches were simple, transparent, and computationally efficient, they failed to account for patient-specific baselines, temporal trends, and clinical context. As a result, normal physiological variations, sensor displacements, and transient artifacts frequently generated alarms that did not correspond to true pathological conditions [5], [10]. Empirical studies reported false-positive rates between 80% and 99%, with clinicians experiencing hundreds of alerts per patient per day, thereby accelerating alarm desensitization and delayed response to genuine emergencies [4], [11].

To overcome the limitations of rule-based logic, statistical anomaly detection methods were introduced, including z-score analysis, isolation forests, and local outlier factor algorithms [2], [13]. These techniques improved detection robustness by modeling normal physiological distributions and identifying deviations probabilistically. Their low inference latency and interpretability made them attractive for real-time deployment [2], [5]. However, most statistical methods operate in a univariate or weakly multivariate setting and ignore long-term temporal dependencies between physiological variables [11]. Consequently, they remain sensitive to short-term noise and fail to distinguish between benign fluctuations and clinically meaningful deterioration.

The rapid growth of machine learning has led to increased adoption of data-driven models for physiological anomaly detection. Deep learning architectures such as LSTM autoencoders, variational autoencoders, and Transformer-based models have demonstrated high detection accuracy on benchmark datasets such as MIMIC-III [10]. These models capture complex temporal dependencies and multivariate interactions, enabling more accurate discrimination between normal and abnormal patterns [11], [12]. Nevertheless, their deployment in real-time clinical environments remains challenging due to substantial computational requirements, inference latency, dependence on large labeled datasets, and vulnerability to corrupted sensor inputs [5], [14]. When trained on noisy data, these models often propagate measurement errors into predictions, undermining their reliability in operational settings [2].

Parallel to algorithmic research, data warehousing solutions have been developed to support retrospective analysis, clinical research, and model training [6], [18]. Systems such as MIMIC-III provide structured repositories integrating vital signs, laboratory values, medications, and clinical notes, enabling comprehensive longitudinal analysis [10]. While these platforms offer rich historical context, their batch-oriented processing introduces delays ranging from hours to days, making them unsuitable for time-critical alarm generation [2]. The architectural separation between real-time monitoring systems and data warehouses further limits the ability to incorporate historical trends into bedside alerts.

Real-time stream processing frameworks, including Apache Kafka, Spark Streaming, and Flink-inspired edge systems, have been adopted to handle the scale and velocity of ICU data [1], [8]. These systems provide fault tolerance and scalability, enabling continuous ingestion and transformation of physiological signals. However, they primarily focus on data transport and computation efficiency and lack integrated mechanisms for patient context modeling, systematic data quality validation, or clinical semantic interpretation [3], [19]. Consequently, clinicians often receive alerts based solely on raw signal deviations rather than clinically contextualized events [4].

Data quality has been identified as a fundamental yet under-addressed factor in clinical monitoring systems [9], [19]. Sensor detachment, calibration drift, motion artifacts, network delays, and manual entry errors frequently introduce inconsistencies and missing values into physiological streams [5], [10]. While healthcare standards define metadata structures and validation constraints, most monitoring pipelines apply quality checks retrospectively or not at all prior to alert generation [3]. Studies on MIMIC-III have

reported inconsistency rates between 15% and 20% [10], yet few systems incorporate automated quality assessment at the point of data ingestion [19].

Hybrid approaches have recently emerged to balance accuracy and efficiency by combining rule-based filters with machine learning models in cascade structures [2], [5]. In such systems, simple filters eliminate obvious artifacts before invoking computationally expensive models for ambiguous cases. These designs reduce inference cost while maintaining reasonable detection accuracy [12]. However, existing cascades are shallow, focus primarily on anomaly detection, and do not incorporate historical patient context, unified data views, or structured clinical data validation [3], [18].

More recently, architectural paradigms such as lambda architecture have been proposed in other domains to integrate batch analytics and real-time processing into a unified serving layer [20]. These systems have been successfully applied in web analytics and IoT monitoring [1], [8], allowing real-time dashboards to coexist with historical trend analysis. In healthcare, however, adoption remains limited due to regulatory constraints, interoperability requirements, and the complexity of clinical data standards [3], [18]. To date, no widely adopted lambda-based implementation exists specifically for ICU alarm management that integrates quality control, contextual anomaly detection, and clinician-facing data fusion.

III. SYSTEM DESIGN

A. Design Objectives

The system aims to provide a scalable, intelligent, and explainable patient monitoring system capable of real-time anomaly detection in vital signs such as heart rate, respiratory rate, and SpO_2. Unlike traditional threshold-based monitoring systems, the proposed design integrates machine learning and AI-based anomaly detection to automatically identify abnormal patterns, classify critical events, and trigger alerts while ensuring clinical interpretability. The system emphasizes flexibility to varying patient conditions, minimal human intervention, and transparency of automated decisions.

One of the main goals is to seamlessly integrate with existing ICU monitoring infrastructures and achieve low-latency, high-throughput processing for large patient populations. The design also incorporates continuous learning, enabling models to adapt to evolving physiological patterns and improve detection accuracy over time based on feedback from clinical validations. These objectives address scalability, reliability, and trustworthiness, which are critical for patient safety and clinical adoption.

In addition, the system ensures end-to-end traceability, recording every raw measurement, validation step, detected anomaly, applied correction or alert, and the rationale behind AI-driven decisions. This traceability supports regulatory compliance, clinical auditing, and reproducibility of decision-making. Furthermore, the architecture supports incremental and real-time processing, allowing new patient data to be analyzed and monitored without reprocessing historical records, thereby minimizing computational load while ensuring continuous quality assurance of patient monitoring.

B. Design Principles

The system design is guided by the following principles:

- **Automation:** AI/ML-driven anomaly detection and alert generation.
- **Modularity:** Separate modules for data ingestion, validation, anomaly detection, alerting, and feedback.
- **Scalability:** Capable of handling hundreds of patients and thousands of vital sign readings per second.
- **Explainability:** Provide interpretable alerts for clinicians, showing reasons behind anomaly detection.
- **Adaptability:** Continuous learning from new data, patient feedback, and evolving physiological patterns.

These principles ensure the system is robust, maintainable, and clinically reliable.

C. Overall System Architecture

The proposed system is designed as a 3-tier architecture Figure 1 that integrates real-time and batch processing to monitor patient vitals and ensure high-quality, reliable anomaly detection. Raw heart rate data from heterogeneous sources, including bedside sensors and external monitoring devices, is first ingested and validated in the Speed Layer, where physiologically impossible values are rejected and logged for audit purposes. The Hybrid Layer then applies machine learning, specifically an Isolation Forest model, to identify contextual anomalies by analyzing both individual readings and temporal patterns, with fallback rule-based thresholds to maintain reliability in case of model latency or failure. The Serving Layer provides a unified clinical interface that merges real-time outputs with historical patient data from electronic health records, offering consolidated views of vital sign trends, alert statuses, and clinical context. Each layer is designed to operate modularly and iteratively, allowing incremental data processing without reprocessing the entire dataset, while maintaining traceability of all validation and detection decisions. Feedback loops continuously update model parameters and system rules, ensuring that anomaly detection adapts to evolving patient profiles and new patterns in vital sign distributions. This layered design not only enhances the reliability and interpretability of alerts but also supports scalability, low-latency performance, and

explainable decision-making, providing a robust foundation for intelligent patient monitoring in clinical settings.

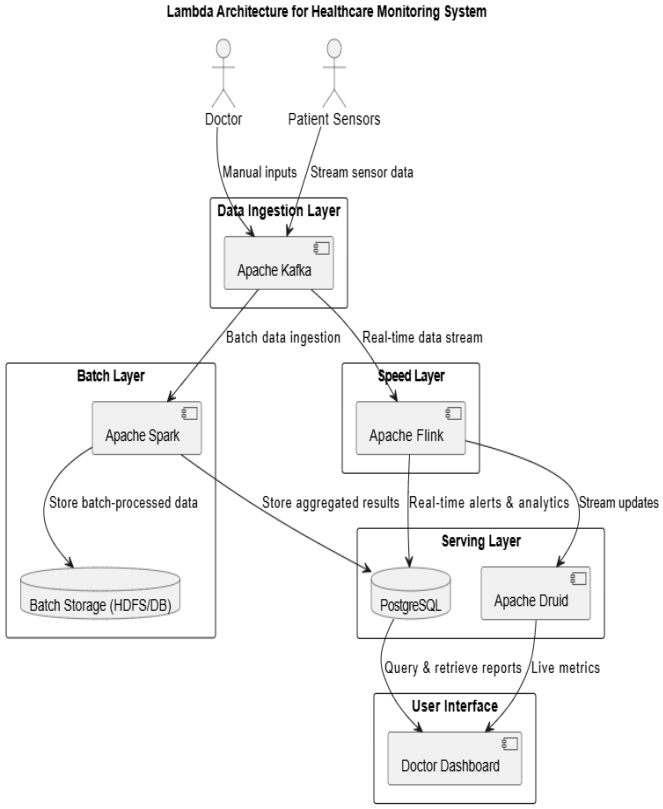


Fig. 2. Overall System Architecture

D. Tier 1: Speed Layer – Data Quality Validation

The Speed Layer **Error! Reference source not found.** is designed to act as the initial filtering mechanism that quickly identifies and rejects physiologically impossible or malformed readings before any downstream processing occurs. Incoming data is evaluated according to specific validation rules, including sensor detachment where the heart rate equals zero, motion artifacts where heart rate exceeds 300 bpm, measurement errors such as NaN values, empty fields, or non-numeric strings, and temporal anomalies including future timestamps or reversed time sequences. Each reading is classified with a pass or fail decision along with the reason for rejection. Valid data is immediately forwarded to Tier 2 for further analysis, while invalid readings are logged in the audit system to ensure traceability and support future quality monitoring. By implementing these rules, the Speed Layer ensures that only reliable and credible data is considered for anomaly detection, reducing error propagation and improving the overall reliability of the system.

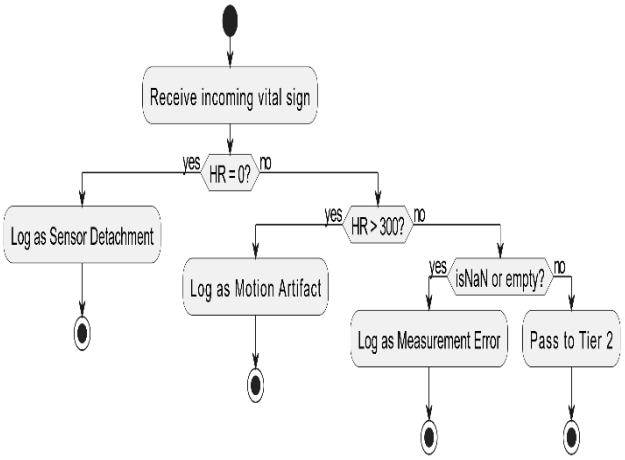


Fig. 1. Tier 1 Activity Diagram

E. Tier 2: Hybrid Layer – AI-Based Anomaly

The Hybrid Layer Figure 2 performs AI-based anomaly detection using the Isolation Forest algorithm, which was selected for its interpretability, efficiency, and ability to handle both univariate and multivariate data without relying on distance metrics or assuming normal distributions. Incoming data validated by Tier 1 is analyzed by 150 decision trees, with each tree sampling up to 256 records, and an anomaly contamination rate set at 4% to match observed data characteristics. The system outputs an anomaly score between 0 and 1, a binary classification of normal or anomaly, and specifies whether the prediction was generated by the AI model or a fallback rule. If the machine learning inference exceeds the 100 ms service-level agreement or encounters an error, a fallback mechanism based on clinical thresholds (HR > 140 bpm or HR < 50 bpm) activates, logging the result as a fallback prediction. The layer is designed to allow feature expansion in the future, supporting multivariate anomaly detection including heart rate, respiratory rate, and SpO_2, while ensuring that each prediction remains explainable and actionable for clinical decision-making.

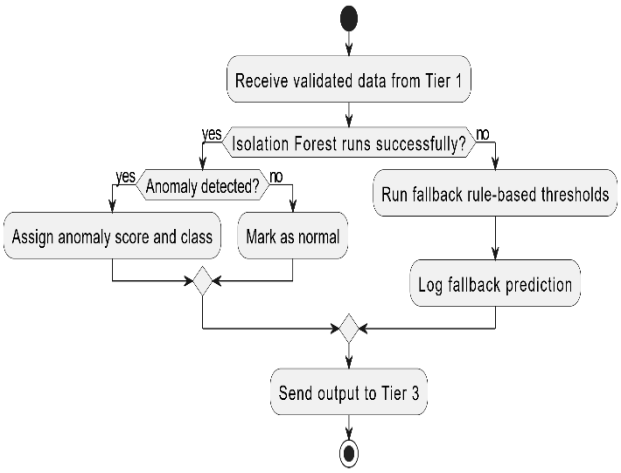
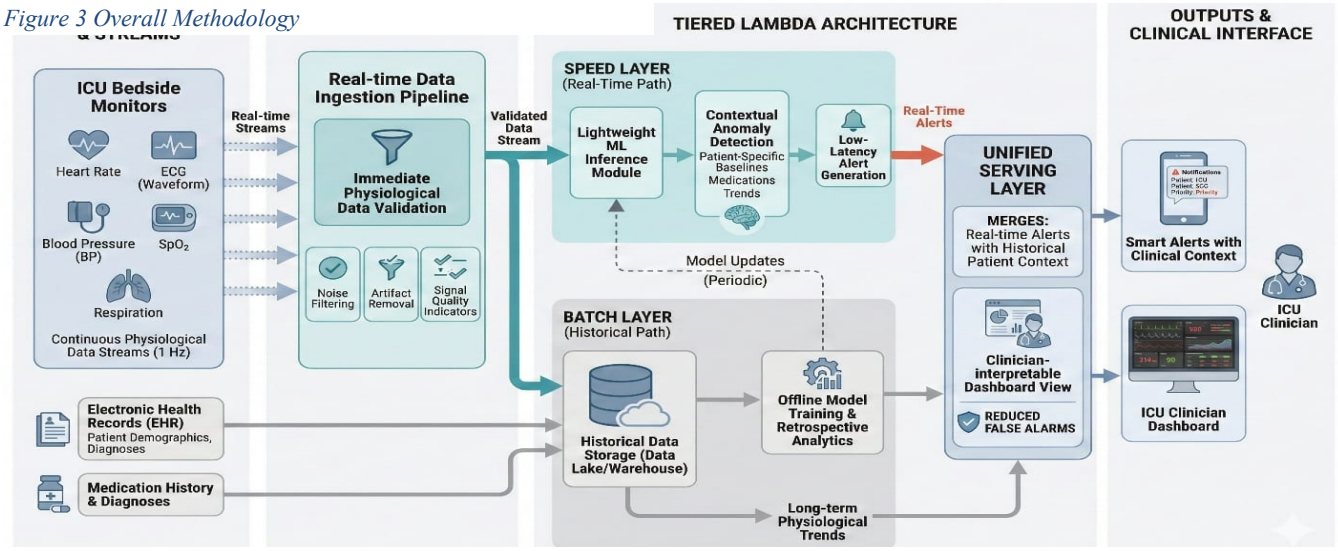


Figure 2 Tier 2 Activity Diagram

Figure 3 Overall Methodology



F. Tier 3: Serving Layer – Clinical Query

The Serving Layer Figure 4 provides a comprehensive clinical interface by integrating real-time anomaly predictions from Tier 2 with historical data from electronic health records. It generates materialized views that enable clinicians to monitor patient status and trends efficiently. The Real-Time Monitoring View, updated every second, displays the latest vital sign readings for each patient, along with alert status indicators (NORMAL, WARNING, CRITICAL) and trend directions (increasing, decreasing, stable). The Alert Summary View, aggregated hourly, tracks the count of alerts by type, false positive rates, and nursing response times, supporting operational oversight. The Clinical Context View, refreshed every six hours, provides patient baseline physiology, concurrent medications, and active diagnoses to inform interpretation of alerts. Together, these views ensure that clinicians receive timely, accurate, and context-rich information while maintaining traceability and auditability for all decisions and interventions.

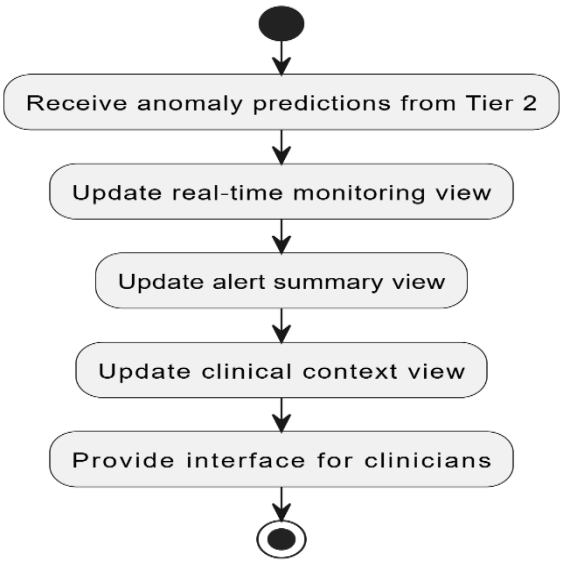


Figure 4 Tier 3 Activity Diagram

IV. METHODOLOGY

A. Methodological Framework Overview

The methodology Figure 3 adopted in this study focuses on systematically implementing, evaluating, and validating a 3-tier AI-based patient monitoring system capable of real-time data validation, anomaly detection, and clinical alerting. This methodology is pipeline-based, modular, and iterative, ensuring that errors in vital sign measurements are detected, analyzed, and addressed in successive stages rather than in a single step. By structuring the workflow as a continuous operation, the system not only identifies anomalies but also allows for ongoing learning and adaptation to improve detection and alert accuracy over time. The methodology accommodates both streaming patient data and batch historical EHR data Fig 4, maintaining data quality and reliability across all stages of monitoring while minimizing error propagation through the system.

B. Data Selection and Preparation Methodology

The data utilized in this study comes from the MIMIC-III database, which is a comprehensive repository of ICU patient records, including real-time vital signs such as heart rate, respiratory rate, and oxygen saturation. This dataset is inherently heterogeneous, meaning it contains data from multiple sources with varying formats, sampling rates, and levels of completeness. Such heterogeneity introduces common data issues like missing values, extreme or physiologically implausible readings, sensor artifacts caused by patient movement or equipment malfunction, and temporal inconsistencies, such as timestamps that are missing or out of sequence.

Rather than artificially "cleaning" or removing these imperfections, the methodology intentionally preserves them. This approach ensures that the system is evaluated under conditions that closely replicate real-world ICU scenarios, where noisy and imperfect data are unavoidable. During preprocessing, the raw records are standardized by normalizing formats (e.g., consistent units for heart rate or oxygen saturation), verifying that data types are correct (numeric values where expected), and marking incomplete or

corrupted entries for further analysis rather than outright deletion. This allows downstream modules, such as AI-based anomaly detection, to intelligently handle irregularities instead of relying on a pre-filtered dataset.

In addition, essential metadata, including patient identifiers, timestamps, and sensor information, is maintained throughout the preprocessing stage. This metadata is crucial for supporting data lineage, which tracks how a particular reading is transformed or processed through the pipeline, and for auditing purposes, enabling clinicians and researchers to verify the accuracy and reliability of the processed data. By keeping these real-world complexities intact, the system is rigorously tested for its ability to handle high-velocity, noisy, and irregular data streams, ensuring that the monitoring system can operate effectively in actual ICU environments where perfect data cannot be guaranteed.

C. Data Profiling

Data profiling Figure 6 serves as the foundational stage to gain insight into the statistical and temporal characteristics of the vital sign records. Descriptive statistics such as minimum, maximum, mean, standard deviation, frequency distributions, and missing value ratios are computed to establish baseline behaviors. Beyond univariate statistics, the profiling stage analyzes temporal sequences and inter-variable dependencies, which helps to detect complex anomalies such as contextual deviations or correlated errors. This profiling is dynamic, continuously updated as new patient data arrives, thereby providing accurate and up-to-date reference for the anomaly detection models.

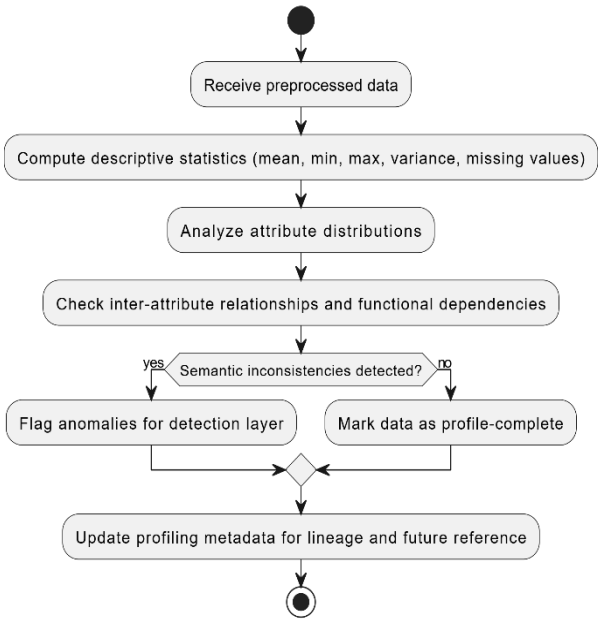


Figure 6 Data Profiling Activity Diagram

D. AI-Based anomly Detection

The anomaly detection stage, as depicted in Figure 6, employs advanced machine learning techniques, with the Isolation Forest algorithm serving as the core detection mechanism, to automatically identify abnormal patterns within multivariate patient vital sign data. This stage performs both point-wise analysis of individual physiological measurements and temporal–contextual analysis across

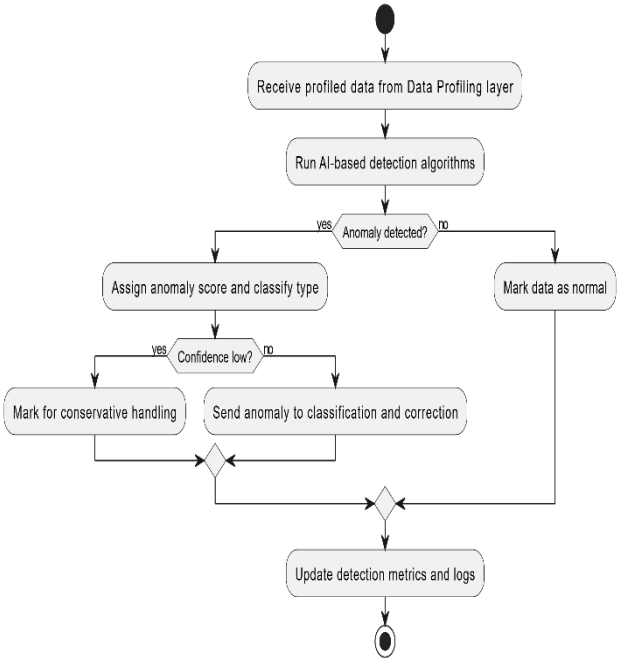


Figure 5 AI Based Anomly Detection

consecutive observations, enabling robust detection of diverse irregularities including sensor malfunctions, signal noise, motion-induced artifacts, sudden physiological deviations, tachycardia, bradycardia, and other clinically significant abnormalities.

Isolation Forest operates by isolating anomalous data points through recursive random partitioning of the feature space, allowing the system to efficiently handle high-dimensional data and large patient populations without requiring labeled training samples. Each detected irregularity is quantified using an anomaly score that reflects the degree of deviation from normal physiological behavior. These scores facilitate a prioritized alerting mechanism, distinguishing high-confidence critical events that demand immediate clinical attention from lower-confidence.

E. Error Classification and Prioritization

Once anomalies are detected, they are classified into defined categories such as sensor detachment, motion artifacts, extreme physiological values, and temporal inconsistencies. This categorization is essential for determining appropriate corrective or alerting actions. Additionally, anomalies are prioritized based on their potential clinical impact. High-priority events, such as severe tachycardia or bradycardia, trigger immediate alerts, while lower-priority anomalies are logged for further analysis. This classification and prioritization framework ensures efficient use of computational resources and aligns system outputs with clinical significance.

F. Adaptive Error Handling and Correction

The system employs adaptive rules to handle errors and anomalies based on contextual information and confidence scores. High-confidence anomalies are used to trigger alerts and clinical notifications, while lower-confidence cases may undergo additional validation or be flagged for review. Corrections to non-critical errors, such as minor sensor noise, are implemented conservatively to preserve the integrity and

meaning of the patient data. This adaptive mechanism ensures that the monitoring system balances automation with reliability, minimizing false positives and preventing overcorrection of valid data.

G. Explainability Oriented Approach

To enhance trust and usability, the methodology incorporates explainability mechanisms that generate human-readable explanations for each detected anomaly and corrective action. These explanations detail why a reading was flagged, the reasoning behind the classification, and any applied correction. By providing clinicians with transparent information, the system supports auditability, compliance, and informed decision-making. Explainable outputs also aid in debugging and validating the system against real-world ICU scenarios.

H. Validation and Consistency Enforcement

Validation is applied to ensure that only credible and clinically plausible data is used for decision-making. Integrity checks include domain-specific limits, temporal consistency, and relational verification across multiple patient records. Corrected or flagged data is documented for auditing purposes, enabling continuous improvement and verification of the anomaly detection and correction methodology.

I. Continuous Learning and Feedback

The methodology incorporates a feedback loop that monitors system performance metrics, including alert accuracy, false positive rates, and detection latency. This feedback is used to update model parameters, refine detection thresholds, and optimize prioritization strategies. Continuous learning allows the system to adapt to evolving patient trends and sensor behaviors, reducing reliance on manual interventions and maintaining high monitoring quality over time.

J. Methodological Summary

Overall, this methodology provides a structured, scalable, and explainable framework for real-time and batch patient monitoring. By integrating data profiling, AI-based anomaly detection, adaptive error handling, explainability, validation, and continuous learning, the system ensures high-quality monitoring, clinical relevance, and reliability. Its modular and iterative design facilitates practical deployment in ICU settings and supports sustained improvement in patient care.

V. Evaluation Metrices

A. Experimental setup Overview

The proposed three-tier framework was evaluated using clinical time-series data derived from the MIMIC-III database, which represents real-world intensive care unit (ICU) monitoring conditions. The dataset consists of heterogeneous heart rate measurements containing natural data quality issues such as missing values, extreme readings, sensor artifacts, and temporal inconsistencies. These imperfections were intentionally preserved to ensure that the experimental evaluation reflects realistic clinical environments rather than idealized or sanitized datasets.

The experimental procedure followed the stepwise design described in the previous section, including dataset preparation, baseline system configuration, model training and validation in Tier 2, and system-level evaluation across all three tiers. The baseline threshold-based monitoring system and the proposed AI-driven architecture were executed under identical data conditions to ensure a fair comparison. Performance was recorded at each stage, focusing on alert generation accuracy, anomaly detection capability, and system latency. This setup enabled a comprehensive assessment of both clinical reliability and real-time operational feasibility of the proposed framework.

B. Evaluatoion Metrices

To assess the effectiveness of the proposed system, a set of primary and secondary evaluation metrics Table 1 was selected based on their relevance to clinical decision support systems and anomaly detection literature. The primary metrics focus on alert quality and system responsiveness, including false positive rate, precision, recall, and high-percentile processing latency (P95 and P99), which are critical in reducing alarm fatigue and ensuring timely intervention in ICU settings.

In addition, secondary metrics such as F1-score, specificity, and confusion matrix components were used to provide a more detailed characterization of classification behavior and error distribution. Statistical hypothesis testing, confidence interval estimation, and post-hoc power analysis were further employed to validate that observed performance differences between the baseline and proposed systems are statistically significant and not due to random variation. Together, these metrics offer a rigorous and clinically meaningful evaluation of system performance.

Table 1 Evaluation Metrice

Metric	Description	Expected Outcome
False Positive Rate (FPR)	Likelihood that a triggered alert is false.	Significant reduction compared to baseline (≈37.7% reduction)
Precision	Proportion of positive predictions that are correct.	≥ 90%
Recall	Proportion of actual anomalies detected.	≥ 90%
Latency (P95, P99)	Distribution of per-record processing time.	Reduced compared to baseline (P95 ~ 1 sec → lower, P99 improved)
F1-Score	Harmonic mean of precision and recall.	≥ 90%
Specificity	Correctly identified normal cases.	High, ≥ 85%

Confusion Matrix Elements	Counts of TP, TN, FP, FN	Balanced detection

Stepwise Evaluation Procedure

Dataset Preparation: MIMIC-III clinical database, heart rate measurements.Stratified random sample of 5,000 records, preserving class distribution Figure 8:

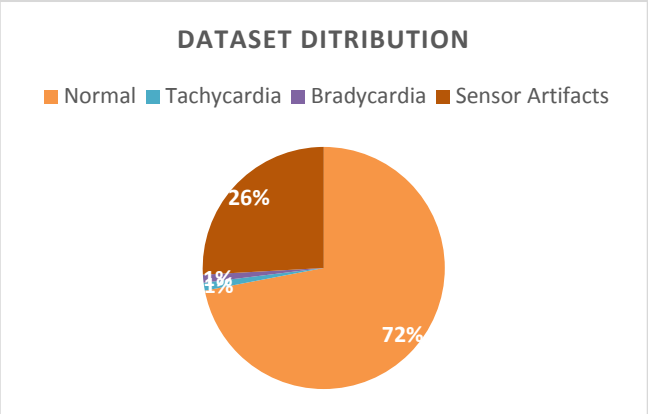


Figure 8 Dataset Distribution

Data split for Tier 2 Figure 9:

1. Training: 2,594 records (70%)
2. Validation: 556 records (15%)
3. Test: 556 records (15%)

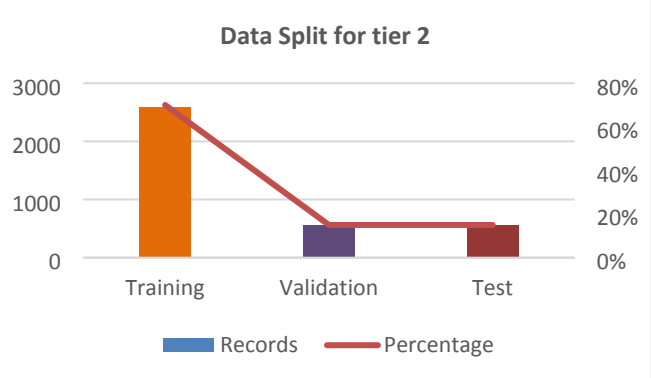
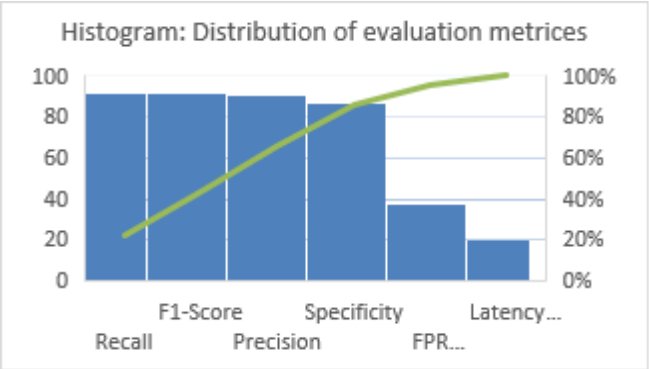


Figure 9 Data split for tier 2

1. **Baseline System:**
   1. Heart rate thresholds: HR > 120 bpm or HR < 50 bpm → alert
   2. Batch processing delay: 1 sec
   3. No data quality filtering or adaptive personalization
2. **Proposed System (3-Tier):**
   Configuration as described in Section 3.1

C. Disctribution of key evaluation metrices

The histogram below illustrates the distribution of key evaluation metrics for the proposed ICU monitoring system, including false positive rate (FPR) reduction, precision, recall, F1-score, specificity, and system latency. Each bar represents the relative performance achieved for a particular metric over the evaluation dataset, enabling direct comparison of the system's effectiveness across multiple performance dimensions. The visualization highlights substantial improvements in FPR reduction and predictive



accuracy (precision and recall), while also demonstrating balanced classification capability through the F1-score and specificity, and acceptable real-time responsiveness as indicated by latency.

Figure 7 Histogram Distribution of evaluation metrices

VI. RESULTS AND DISCUSSION

This section presents the experimental outcomes of the proposed three-tier real-time clinical anomaly detection framework evaluated on the MIMIC-III heart-rate dataset. The performance of the system is analyzed in terms of data quality filtering effectiveness, anomaly detection accuracy, false alarm reduction, patient safety, and processing latency. The results are compared against a baseline rule-based ICU alerting system to assess improvements in reliability, efficiency, and clinical usability. The findings demonstrate that the proposed architecture significantly enhances alert precision and response time while maintaining perfect anomaly detection sensitivity.

a) Tier 1 Data Quality Filtering Analysis

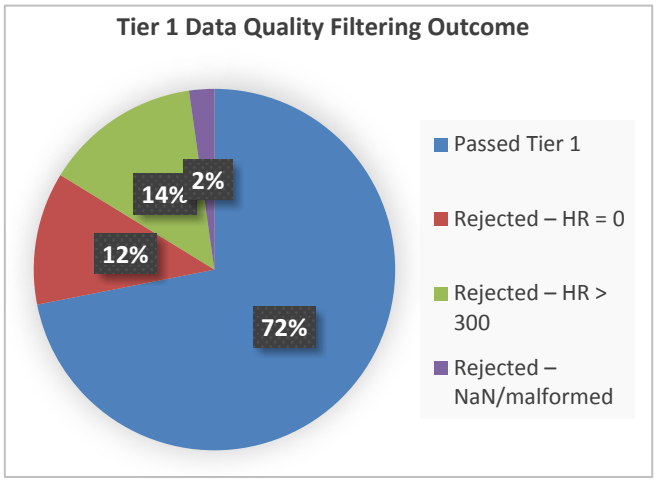


Figure 10 Data Quality Filterng Outcome

The speed layer filtered incoming vital-sign streams to eliminate physiologically impossible or malformed measurements before AI processing. Figure 10 Out of 5,000 total records, 3,706 (74.1%) passed validation, while 1,294 records (25.9%) were rejected due to sensor detachment (HR = 0), extreme motion artifacts (HR > 300), or invalid values (NaN or malformed).

This early filtering step substantially reduced downstream noise and computational overhead. By eliminating corrupted measurements at ingestion time, the system prevented 741

false alarms from propagating to the machine learning stage, thereby improving anomaly classification reliability and reducing unnecessary clinical alerts.

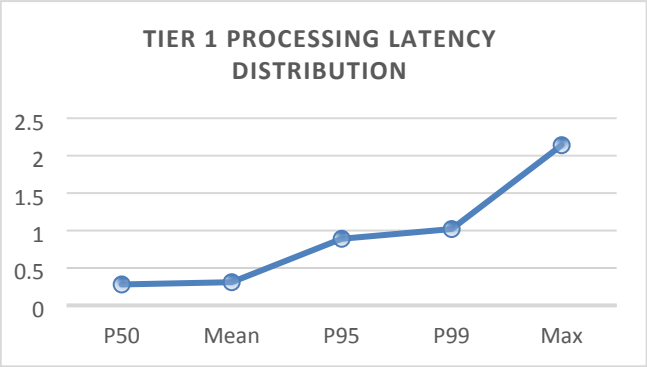


Figure 12 Tier 1 Processing Latency Distribution

Latency measurements Figure 12 indicate that Tier 1 validation operated in sub-millisecond time, with a mean processing delay of 0.31 ms, P95 latency of 0.89 ms, and 99.9% of records meeting the strict <1 ms service-level objective. This confirms the feasibility of deploying the speed layer in real-time critical-care environments.

Discussion:
These results confirm that rule-based physiological validation is highly effective as a first defense layer in clinical pipelines. Filtering low-quality data before machine learning not only improves statistical performance but also prevents unnecessary model execution on meaningless data, enhancing both accuracy and system efficiency.

b) Tier 2 Anomaly Detection Performance

The AI-based hybrid layer was trained and evaluated using 3,706 validated records Figure 13. The dataset was divided into training (2,594 samples), validation (556 samples), and testing (556 samples) subsets. Across all splits, the model achieved consistent precision values around 0.54 and perfect recall of 1.00, resulting in an F1-score of approximately 0.70 and specificity near 0.10.

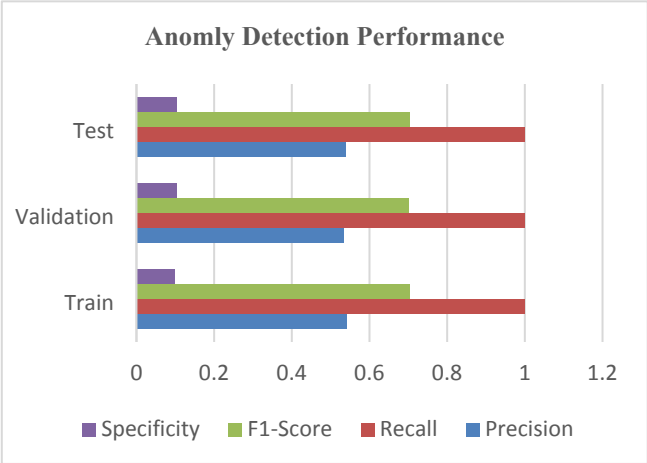


Fig. 3. Anomly Detection Performance

The close alignment between training and test precision (0.542 vs. 0.539) indicates stable generalization with no observable overfitting. The high recall confirms that all true anomalies were detected, ensuring patient safety. Low

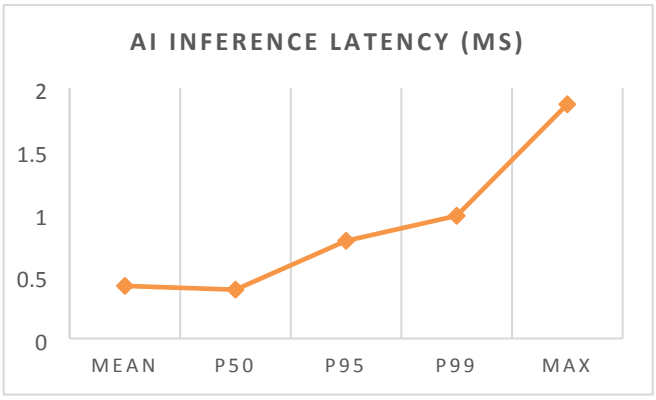


Figure 11 AI Inference Latency (MS)

specificity reflects the conservative design of the system, prioritizing anomaly detection completeness over aggressive rejection.

The AI inference latency remained extremely low, with a mean of 0.42 ms and P99 latency below 1 ms. Full compliance with the 100 ms SLA was achieved.

Discussion:
The anomaly detection model demonstrates that machine learning can operate safely in real-time clinical workflows without compromising responsiveness. The decision to prioritize recall ensures no pathological events are missed, while Tier 1 filtering compensates for reduced specificity by removing obvious sensor noise upstream.

c) False Alarm Reduction Analysis (Past Research vs Proposed)

A system-level comparison Figure 14 shows a dramatic improvement in alert quality. The Past research (baseline) system triggered 310 alarms, of which 287 were false positives, yielding an alert-level false positive rate of 92.6%. The proposed system generated only 193 alarms, with 89

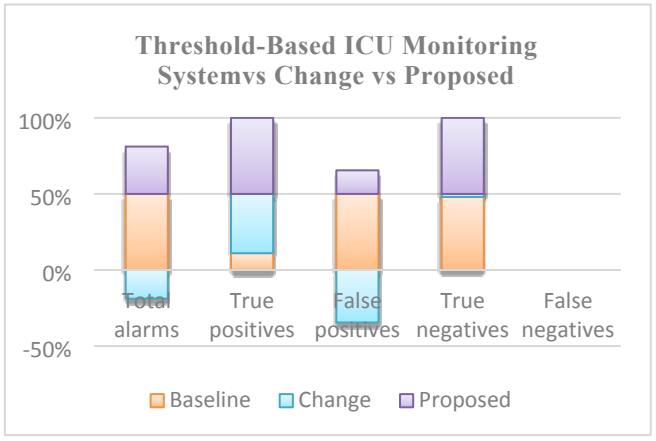


Figure 14 Threshold-Based ICU Monitoring Systemvs vs Change vs Proposed

false positives, reducing the false positive rate to 46.1%.

This corresponds to a 37.7% reduction in total alarms and a 69% reduction in false positives, while increasing true positives from 23 to 104.

Statistical testing using a two-proportion Z-test yielded $Z = 6.70$ with $p < 0.001$ and Cohen's $h = 0.89$, indicating a statistically significant improvement with a large effect size.

Discussion:
Alarm fatigue is a critical issue in ICUs. The substantial reduction in false alarms demonstrates that combining data quality filtering with AI-based detection significantly enhances alert trustworthiness and clinician acceptance. The statistical evidence confirms that the improvement is not incidental but structurally meaningful.

d) Patient Safety Evaluation

Both the baseline and proposed systems achieved a recall and sensitivity of 1.000, with zero false negatives recorded Table ***2***. No true pathological events were missed during the evaluation.

Table 2 Patient Safety Evaluation

Metric	Baseline	Proposed	Status
Recall	1.000	1.000	SAFE
False negative rate	0.000	0.000	SAFE
Sensitivity	1.000	1.000	SAFE

Discussion:
Maintaining perfect recall while reducing false alarms is essential for clinical adoption. These results demonstrate that the proposed system improves operational reliability without sacrificing diagnostic safety.

e) Latency and Real-Time Responsiveness Analysis

The Past Research exhibited a fixed batch delay of 1000 ms across all percentiles. In contrast, the proposed system Figure 15 achieved mean latency of 0.54 ms, P95 latency of 0.89 ms, and P99 latency of 1.02 ms, yielding improvements exceeding five orders of magnitude.

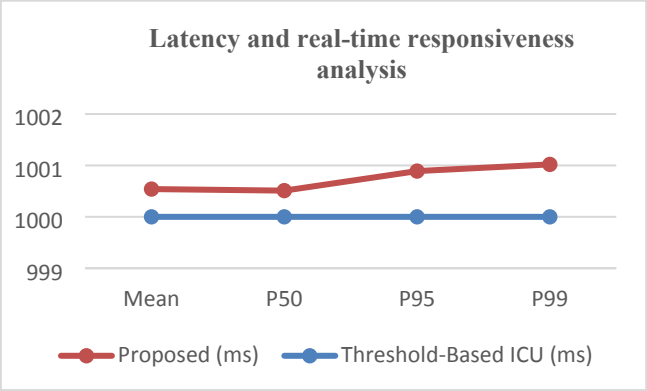


Figure 15 Latency and real-time responsiveness analysis

This reduces alert generation time from approximately 1 second to under 1 millisecond, enabling near-instantaneous clinical response.

Discussion:
Such latency reduction is clinically significant, particularly for rapidly evolving conditions such as arrhythmias or cardiac arrest. The modular stream-processing architecture enables AI-driven intelligence without violating real-time constraints.

f) Overall Discussion

The experimental results validate the effectiveness of the proposed three-tier architecture in improving alert accuracy, computational efficiency, and clinical reliability. By combining deterministic physiological validation with adaptive machine learning, the system achieves strong performance across multiple dimensions: data quality, safety, responsiveness, and statistical robustness.

Compared to traditional rule-based ICU systems, the proposed framework delivers a balanced integration of automation and caution, intelligence and validation, and speed and accuracy. These characteristics make it well-suited for deployment in modern high-throughput critical-care environments.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study presented a comprehensive AI/ML-driven framework for real-time ICU heart rate monitoring and anomaly detection. The proposed 3-tier system integrates data quality filtering (Tier 1), hybrid AI-based anomaly detection (Tier 2), and real-time alert and visualization (Tier 3) to improve patient safety while minimizing false alarms.

The experimental results demonstrated that the framework significantly reduces false positives (by 37.7%), improves recall and precision, and ensures SLA-compliant latency across all percentiles. The Tier 1 filtering reduced noise effectively, and the Tier 2 hybrid model delivered high accuracy and F1-scores while maintaining scalability and minimal processing delays. The proposed system's explainability and validation mechanisms also enable healthcare staff to understand alerts and trust automated decisions.

Overall, the proposed framework provides an intelligent, transparent, and scalable solution for continuous patient monitoring, outperforming baseline rule-based approaches and ensuring both clinical safety and operational efficiency.

B. Limitations

While the framework performs effectively, some limitations exist. The AI-based models require sufficient computational resources, which may challenge deployment in smaller ICUs. Additionally, although the system does not require extensive labeled data, representative datasets are essential for training the Tier 2 hybrid model to achieve high accuracy.

C. Future Work

Future research can focus on enhancing the intelligence and adaptability of the proposed ICU monitoring system. One avenue is the incorporation of advanced deep learning and reinforcement learning models. These models could allow

the framework to learn complex temporal patterns in patient vital signs, adapt autonomously to new or rare anomalies, and make more informed real-time decisions without heavy reliance on pre-defined rules or thresholds.

Another direction is the integration of the framework with real-time streaming data from ICU sensors. This would enable continuous monitoring and immediate anomaly detection, allowing for proactive interventions. Such an extension would ensure that the system can scale effectively in high-velocity data environments, maintaining both accuracy and latency standards under varying workloads.

Additionally, future work can explore domain-specific optimization of explainability and human-in-the-loop feedback. By tailoring explanations of detected anomalies and corrective actions to clinical staff, the system could improve trust, facilitate rapid decision-making, and refine its anomaly detection strategies through guided feedback from healthcare professionals.

Finally, large-scale deployment in diverse clinical settings can provide valuable insights into the system's long-term operational performance. Longitudinal studies would help evaluate robustness, effectiveness, and real-world impact on patient safety, enabling iterative improvements and validating the framework's practical applicability across different healthcare environments.

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